



40.0.1 ISRO CSE 2023 | Question: 1



Consider the context-free grammar G below for arithmetic expressions :

$$E \rightarrow E + E | E \times E | \text{id}$$

Which of the following statements is TRUE:

- A. The string " id + id × id " has no parse tree according to G
- B. The string " id + id × id " has only one parse tree according to G
- C. The string " id + id × id " has exactly two parse trees according to G
- D. The string " id + id × id " has more than two parse trees according to G

isro-cse-2023 theory-of-computation ambiguous-grammar

Answer key

40.1

Closure Property (2)

40.1.1 Closure Property: ISRO CSE 2017 | Question: 77



If L and P are two recursively enumerable languages then they are not closed under

- A. Kleene star L^* of L
- B. Intersection $L \cap P$
- C. Union $L \cup P$
- D. Set difference

isro2017 set-theory theory-of-computation recursive-and-recursively-enumerable-languages closure-property

Answer key

40.1.2 Closure Property: ISRO CSE 2018 | Question: 25



CFG (Context Free Grammar) is not closed under:

- A. Union
- B. Complementation
- C. Kleene star
- D. Product

isro2018 closure-property context-free-language theory-of-computation

Answer key

40.2

Context Free Grammar (5)

40.2.1 Context Free Grammar: ISRO CSE 2018 | Question: 27



A CFG (Context Free Grammar) is said to be in Chomsky Normal Form (CNF), if all the productions are of the form $A \rightarrow BC$ or $A \rightarrow a$. Let G be a CFG in CNF . To derive a string of terminals of length x , the number of products to be used is:

- a. $2x - 1$
- b. $2x$
- c. $2x + 1$
- d. 2^x

isro2018 context-free-grammar theory-of-computation

Answer key

40.2.2 Context Free Grammar: ISRO CSE 2020 | Question: 39



The language which is generated by the grammar $S \rightarrow aSa \mid bSb \mid a \mid b$ over the alphabet of $\{a, b\}$ is the set of

- A. Strings that begin and end with the same symbol
- B. All odd and even length palindromes
- C. All odd length palindromes
- D. All even length palindromes

isro-2020 theory-of-computation context-free-grammar normal

Answer key

40.2.3 Context Free Grammar: ISRO CSE 2023 | Question: 34



Consider the context-free grammar G below. Here S is the starting non-terminal symbol, while a and b are terminal symbols.

$$\begin{aligned} S &\rightarrow aSb \mid T \\ T &\rightarrow bT \mid b \end{aligned}$$

The language generated by G is:

- A. $\{a^m b^n : 0 \leq m < n\}$
- B. $\{a^m b^n : 0 < m < n\}$
- C. $\{a^m b^n : 0 < m \leq n\}$
- D. $\{a^n b^n : n \geq 0\}$

isro-cse-2023 context-free-grammar grammar theory-of-computation

Answer key

40.2.4 Context Free Grammar: ISRO-DEC2017-22



Consider the grammar with productions

$$S \rightarrow aSb \mid SS \mid \varepsilon$$

This grammar is

- A. not context-free, not linear
- B. not context-free, linear
- C. context-free, not linear
- D. context-free, linear

isrodec2017 theory-of-computation grammar context-free-grammar

Answer key

40.2.5 Context Free Grammar: ISRO-DEC2017-23



Identify the language generated by the following grammar

$$S \rightarrow AB$$

$$A \rightarrow aAb \mid \varepsilon$$

$$B \rightarrow bB \mid b$$

- A. $\{a^m b^n \mid n \geq m, m > 0\}$
- B. $\{a^m b^n \mid n \geq m, m \geq 0\}$
- C. $\{a^m b^n \mid n > m, m > 0\}$
- D. $\{a^m b^m \mid n > m, m \geq 0\}$

isrodec2017 theory-of-computation context-free-grammar

Answer key

40.3.1 Context Free Language: GATE CSE 2009 | Question: 12, ISRO2016-37



$$S \rightarrow aSa \mid bSb \mid a \mid b$$

The language generated by the above grammar over the alphabet $\{a, b\}$ is the set of:

- A. all palindromes
B. all odd length palindromes
C. strings that begin and end with the same symbol
D. all even length palindromes

gatecse-2009 theory-of-computation context-free-language easy isro2016

Answer key

40.3.2 Context Free Language: ISRO CSE 2020 | Question: 37



Context free languages are closed under

- A. union, intersection
B. union, kleene closure
C. intersection, complement
D. complement, kleene closure

isro-2020 theory-of-computation context-free-language easy

Answer key

40.3.3 Context Free Language: ISRO CSE 2023 | Question: 56



Which of the following statements is FALSE?

- A. The intersection of a regular language and a context-free language is context-free
B. The intersection of a regular language and context-free language is regular
C. The union of two context-free languages is context-free
D. The union of two regular languages is regular

isro-cse-2023 theory-of-computation context-free-language regular-language

Answer key

40.3.4 Context Free Language: ISRO-DEC2017-26



Which of the following are context-free?

$$A = \{a^n b^n a^m b^m \mid m, n \geq 0\}$$
$$B = \{a^m b^n a^m b^n \mid m, n \geq 0\}$$
$$C = \{a^m b^n \mid m \neq 2n, m, n \geq 0\}$$

- A. A and B only
B. A and C only
C. B and C only
D. C only

isrodec2017 theory-of-computation context-free-language

Answer key

40.4

Context Sensitive (1)

40.4.1 Context Sensitive: ISRO CSE 2017 | Question: 11



Given the following statements

- S1 : Every context-sensitive language L is recursive
- S2 : There exists a recursive language that is not context-sensitive

Which statements are true?

- A. Only S1 is correct
- C. Both S1 and S2 are not correct

- B. Only S2 is correct
- D. Both S1 and S2 are correct

isro2017 theory-of-computation context-sensitive

Answer key

40.5

Finite Automata (5)

40.5.1 Finite Automata: ISRO CSE 2014 | Question: 2



The number of states required by a Finite State Machine, to simulate the behavior of a computer with a memory capable of storing 'm' words, each of length 'n' bits is?

- A. $m \times 2^n$
- B. 2^{m+n}
- C. 2^{mn}
- D. $m + n$

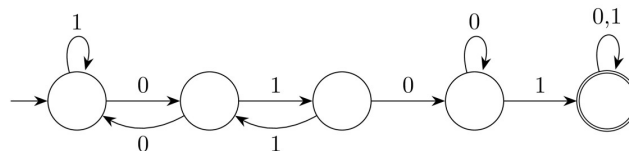
theory-of-computation finite-automata isro2014

Answer key

40.5.2 Finite Automata: ISRO CSE 2014 | Question: 79



Consider the following Deterministic Finite Automaton M .



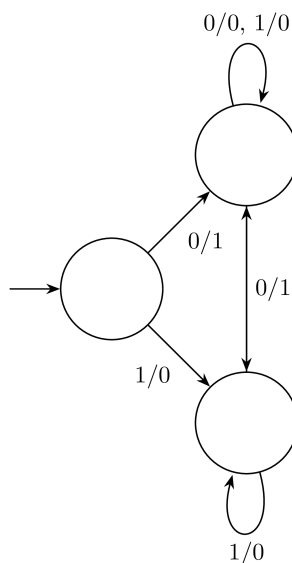
Let S denote the set of eight length bit strings whose second, third, sixth and seventh bits are 1. The number of strings in S that are accepted by M is

- A. 0
- B. 1
- C. 2
- D. 3

isro2014 theory-of-computation finite-automata

Answer key

40.5.3 Finite Automata: ISRO CSE 2018 | Question: 26



The FSM (Finite State Machine) machine pictured in the figure above

- A. Complements a given bit pattern
C. Increments a given bit pattern by 1

- B. Finds 2's complement of a given bit pattern
D. Changes the sign bit

isro2018 finite-automata theory-of-computation

Answer key

40.5.4 Finite Automata: ISRO CSE 2020 | Question: 41

Minimum number of states required in DFA accepting binary strings not ending in "101" is

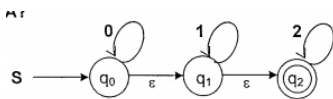
- A. 3 B. 4 C. 5 D. 6

isro-2020 theory-of-computation finite-automata normal

Answer key

40.5.5 Finite Automata: ISRO-2013-73

What are the final states of the DFA generated from the following NFA?



- A. q_0, q_1, q_2
C. $q_0, [q_1, q_2]$
B. $[q_0, q_1], [q_0, q_2], []$
D. $[q_0, q_1], q_2$

isro2013 theory-of-computation finite-automata

Answer key

40.6 First Order Logic (1)

40.6.1 First Order Logic: ISRO-DEC2017-7

If $T(x)$ denotes x is a trigonometric function, $P(x)$ denotes x is a periodic function and $C(x)$ denotes x is a continuous function then the statement "It is not the case that some trigonometric functions are not periodic" can be logically represented as

- A. $\neg \exists x [T(x) \wedge \neg P(x)]$
B. $\neg \exists x [T(x) \vee \neg P(x)]$
C. $\neg \exists x [\neg T(x) \wedge \neg P(x)]$
D. $\neg \exists x [T(x) \wedge P(x)]$

isrodec2017 propositional-logic first-order-logic logical-reasoning

Answer key

40.7 Grammar (3)

40.7.1 Grammar: ISRO CSE 2008 | Question: 7

Consider the grammar

- $S \rightarrow ABCc \mid bc$
- $BA \rightarrow AB$
- $Bb \rightarrow bb$
- $Ab \rightarrow ab$

- $Aa \rightarrow aa$

Which of the following sentences can be derived by this grammar?

- A. abc B. aab C. abcc D. abbc

isro2008 theory-of-computation context-free-language grammar

Answer key 

40.7.2 Grammar: ISRO CSE 2011 | Question: 68



Which of the following sentences can be generated by

$S \rightarrow aS \mid bA$

$A \rightarrow d \mid cA$

- A. bccdd B. abbcca C. abcabc D. abcd

isro2011 theory-of-computation context-free-language grammar

Answer key 

40.7.3 Grammar: ISRO CSE 2023 | Question: 77



Consider the context-free grammar G below. There S is the starting non terminal symbol, while a and b are terminal symbols.

$S \rightarrow aaSb \mid T$

$T \rightarrow Tb \mid a$

Which of the following statements is true about the language $L(G)$ generated by G ?

- A. $aabbaabb$ belongs to $L(G)$ but $aabb$ does not
 B. $aaaaabbb$ belongs to $L(G)$ but $aaaabb$ does not
 C. $aaaabb$ belongs to $L(G)$ but $aabbaabb$ does not
 D. $aaabb$ belongs to $L(G)$ but $aaaaabbb$ does not

isro-cse-2023 theory-of-computation grammar

Answer key 

40.8 Identify Class Language (3)

40.8.1 Identify Class Language: ISRO CSE 2016 | Question: 38



What is the highest type number that can be assigned to the following grammar?

$S \rightarrow Aa, A \rightarrow Ba, B \rightarrow abc$

- A. Type 0 B. Type 1 C. Type 2 D. Type 3

theory-of-computation identify-class-language isro2016

Answer key 

40.8.2 Identify Class Language: ISRO CSE 2020 | Question: 40



Which of the following classes of languages can validate an IPv4 address in dotted decimal format? It is to be ensured that the decimal values lie between 0 and 255.

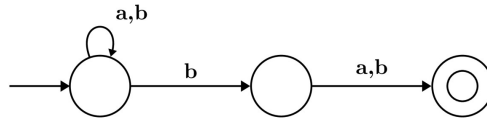
- A. RE and higher B. CFG and higher
 C. CSG and higher D. Recursively enumerable language

Answer key

40.8.3 Identify Class Language: ISRO CSE 2023 | Question: 33



Consider the Nondeterministic Finite State Automaton (NFA) below. State with a small incoming arrow are initial states while a double circle state denotes a final state.



The number of states in the **MINIMAL** Deterministic Finite-State Automaton (DFA) for this language is:

- A. 2 B. 4 C. 6 D. 8

Answer key

40.9 Minimal State Automata (1)

40.9.1 Minimal State Automata: ISRO CSE 2014 | Question: 12



How many states are there in a minimum state deterministic finite automaton accepting the language $L = \{w \mid w \in \{0,1\}^*, \text{ number of 0's is divisible by 2 and number of 1's is divisible by 5, respectively } \}$?

- A. 7 B. 9 C. 10 D. 11

Answer key

40.10 Pushdown Automata (1)

40.10.1 Pushdown Automata: GATE CSE 2009 | Question: 16, ISRO2017-12



Which one of the following is FALSE?

- A. There is a unique minimal DFA for every regular language
 B. Every NFA can be converted to an equivalent PDA.
 C. Complement of every context-free language is recursive.
 D. Every nondeterministic PDA can be converted to an equivalent deterministic PDA.

Answer key

40.11 Recursive and Recursively Enumerable Languages (3)

40.11.1 Recursive and Recursively Enumerable Languages: GATE CSE 2008 | Question: 13, ISRO2016-36



If L and $\bar{L}$ are recursively enumerable then L is

- A. regular B. context-free

C. context-sensitive

D. recursive

gatecse-2008 theory-of-computation easy isro2016 recursive-and-recursively-enumerable-languages

Answer key

40.11.2 Recursive and Recursively Enumerable Languages: ISRO CSE 2011 | Question: 79



A problem whose language is recursion is called?

A. Unified problem

B. Boolean function

C. Recursive problem

D. Decidable

isro2011 theory-of-computation recursive-and-recursively-enumerable-languages

Answer key

40.11.3 Recursive and Recursively Enumerable Languages: ISRO CSE 2023 | Question: 55



Which of the following statements is NOT true?

A. If a language is recursive its complement is recursive

B. If a language is recursive its complement is recursively enumerable

C. If a language and its complement are recursively enumerable it is recursive

D. If a language is recursively enumerable its complement is also recursively enumerable

isro-cse-2023 theory-of-computation decidability recursive-and-recursively-enumerable-languages

Answer key

40.12 Recursively Enumerable Languages (1)

40.12.1 Recursively Enumerable Languages: ISRO-DEC2017-24



Let L_1 be regular language, L_2 be a deterministic context free language and L_3 a **recursively enumerable language**, but not recursive. Which one of the following statements is false ?

A. $L_3 \cap L_1$ is recursive

B. $L_1 \cap L_2 \cap L_3$ is recursively enumerable

C. $L_1 \cup L_2$ is context free

D. $L_1 \cap L_2$ is context free

isrodec2017 theory-of-computation regular-language context-free-language recursively-enumerable-languages

Answer key

40.13 Regular Expression (5)

40.13.1 Regular Expression: GATE CSE 1995 | Question: 1.9 , ISRO2017-13



In some programming language, an identifier is permitted to be a letter followed by any number of letters or digits. If L and D denote the sets of letters and digits respectively, which of the following expressions defines an identifier?

A. $(L + D)^+$

B. $(L.D)^*$

C. $L(L + D)^*$

D. $L(L.D)^*$

gate1995 theory-of-computation regular-expression easy isro2017

Answer key

40.13.2 Regular Expression: ISRO CSE 2016 | Question: 33



Let $L = \{w \in (0 + 1)^* \mid w \text{ has even number of 1's}\}$, i.e. L is the set of all bit strings with even number of 1's. Which one of the regular expression below represents L ?

- A. $(0^*10^*1)^*$
 C. $0^*(10^*1^*)^*0^*$

- B. $0^*(10^*10^*)^*$
 D. $0^*1(10^*1)^*10^*$

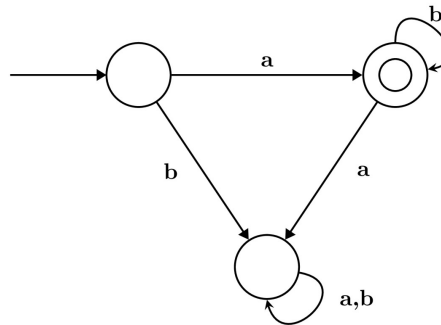
theory-of-computation regular-expression isro2016

Answer key

40.13.3 Regular Expression: ISRO CSE 2023 | Question: 13



Consider the deterministic finite-state automaton (DFA) below. The alphabet is $\{a, b\}$. The state with a small incoming arrow is the initial state, while the double circle state denotes a final state.



Which of the following regular expressions defines the language accepted by the DFA?

- A. ab^* B. a^*b^* C. $(ab)^*$ D. a^*b

isro-cse-2023 theory-of-computation regular-expression

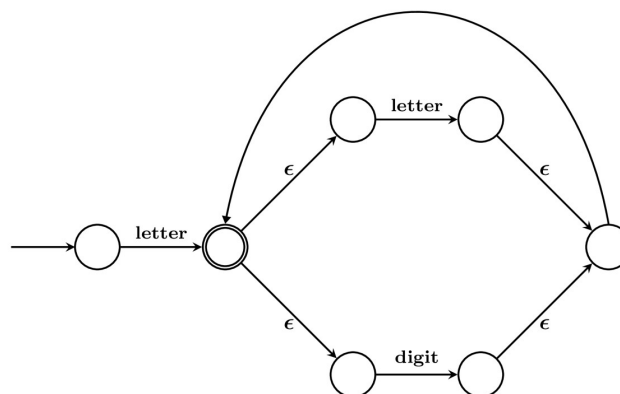
Answer key

40.13.4 Regular Expression: ISRO CSE 2023 | Question: 64



Consider the NFA with epsilon-transitions below, which is meant to accept strings corresponding to a lexical token id. Here the token digit corresponds to the extended regular expression $[0-9]$ and letter corresponds to $[a-zA-Z]$. In the diagram, a small incoming edge indicates a start state while a double circle indicates a final state.

Which of the following extended regular expressions corresponds to the token id?



- A. $\text{letter (letter | digit)}^*$ B. $(\text{letter | digit})^* \text{letter}$
 C. $(\text{letter | digit})^*$ D. $\text{letter (letter}^* \text{ | digit}^*)$

Answer key 

40.13.5 Regular Expression: ISRO-DEC2017-21



For $\Sigma = \{a, b\}$ the regular expression $r = (aa)^*(bb)^*b$ denotes

- A. Set of strings with 2 a 's and 2 b 's
- B. Set of strings with 2 a 's 2 b 's followed by b
- C. Set of strings with 2 a 's followed by b 's which is a multiple of 3
- D. Set of strings with even number of a 's followed by odd number of b 's

Answer key 

40.14

Regular Language (3)

40.14.1 Regular Language: ISRO CSE 2018 | Question: 24



Choose the correct statement -

- a. $A = \{a^n b^n \mid n = 1, 2, 3, \dots\}$ is a regular language
- b. The set B , consisting of all strings made up of only a 's and b 's having equal number of a 's and b 's defines a regular language
- c. $L(A^*B) \cap B$ gives the set A
- d. None of the above

Answer key 

40.14.2 Regular Language: ISRO CSE 2020 | Question: 38



Which of the following is true?

- A. Every subset of a regular set is regular
- B. Every finite subset of non-regular set is regular
- C. The union of two non regular set is not regular
- D. Infinite union of finite set is regular

Answer key 

40.14.3 Regular Language: ISRO2015-43



Let R_1 and R_2 be regular sets defined over the alphabet, then

- A. $R_1 \cap R_2$ is not regular
- B. $R_1 \cup R_2$ is not regular
- C. $\Sigma^* - R_1$ is regular
- D. R_1^* is not regular

Answer key 

40.15

Strings (1)

40.15.1 Strings: ISRO CSE 2023 | Question: 73



From a character string of length m , the number of sub-strings of all lengths that can be formed are:

- A. m^2
- B. m
- C. $m(m+1)/2$
- D. $m(\log m)$

isro-cse-2023 counting strings

Answer key

40.16

Turing Machine (3)

40.16.1 Turing Machine: ISRO CSE 2014 | Question: 59



Which of the following is FALSE with respect to possible outcomes of executing a Turing Machine over a given input?

- A. it may halt and accept the input
- B. it may halt by changing the input
- C. it may halt and reject the input
- D. it may never halt

isro2014 theory-of-computation turing-machine

Answer key

40.16.2 Turing Machine: ISRO CSE 2016 | Question: 32



AN FSM(finite state machine) can be considered to be a turing machine of finite tape length

- A. without rewinding capability and unidirectional tape movement
- B. rewinding capability and unidirectional tape movement
- C. without rewinding capability and bidirectional tape movement
- D. rewinding capability and bidirectional tape movement

theory-of-computation isro2016 turing-machine

Answer key

40.16.3 Turing Machine: ISRO-DEC2017-25



Let $L = \{a^p \mid p \text{ is a prime}\}$. Then which of the following is true

- A. It is not accepted by a Turing Machine
- B. It is regular but not context-free
- C. It is context-free but not regular
- D. It is neither regular nor context-free ,but accepted by a Turing Machine

isrodec2017 theory-of-computation regular-language context-free-language turing-machine

Answer key

Answer Keys

40.0.1	C	40.1.1	D	40.1.2	B	40.2.1	A	40.2.2	C
40.2.3	A	40.2.4	C	40.2.5	X	40.3.1	B	40.3.2	B
40.3.3	B	40.3.4	B	40.4.1	D	40.5.1	C	40.5.2	C

40.5.3	X
40.7.2	D
40.9.1	C
40.12.1	A
40.13.5	D
40.16.1	B

40.5.4	B
40.7.3	B
40.10.1	D
40.13.1	C
40.14.1	D
40.16.2	A

40.5.5	A
40.8.1	D
40.11.1	D
40.13.2	B
40.14.2	B
40.16.3	D

40.6.1	A
40.8.2	A
40.11.2	D
40.13.3	A
40.14.3	C

40.7.1	X
40.8.3	B
40.11.3	D
40.13.4	A
40.15.1	C